Mannheim Business School

Assignment

Defining Markets for plastic waste collection business model

Master in Sustainability and Impact Management

Module - Sustainable Entrepreneurship

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Declaration of Authorship

(Course: *MSIM25*)

”I hereby declare that the paper presented “*Defining Markets for plastic waste collection business model*” is my own work and that I have not called upon the help of a third party. In addition, I affirm that neither I nor anybody else has submitted this paper or parts of it to obtain credits elsewhere before. I have clearly marked and acknowledged all quotations or references that have been taken from the works of others. All secondary literature and other sources are marked and listed in the bibliography. The same applies to all charts, diagrams and illustrations as well as to all Internet resources. Moreover, I consent to my paper being electronically stored and sent anonymously in order to be checked for plagiarism.”

Michael Spilger

Mannheim, May 3, 2025

Problem Statement: Plastic waste has reached crisis levels globally. Between 1950 and 2015, 8.3 billion tonnes of plastic were produced, generating 6.3 billion tonnes of waste, with only 9% recycled(Geyer, Jambeck, and Law 2017). The rest pollutes land and oceans. Only 14% of plastic packaging is recycled, causing $80–120 billion in annual losses(Ezeudu, Tenebe, and Ujah 2024). Developing regions face greater challenges: LAC countries recycle less than 10% of plastic waste(United Nations Environment Programme 2023), with 17,000 tonnes trashed daily in open dumps(Husaini et al. 2024).

Business Idea: **Money for Waste** is an IoT-enabled, gamified recycling initiative addressing plastic pollution and wasted value. The startup uses smart bins and a mobile app to incentivize sorting and depositing plastic waste. Users earn points and rewards, turning recycling into a game-like experience. IoT tracks materials in real time, and sorted plastics are sold to recyclers, generating revenue and closing the loop. By rewarding consumers, *Money for Waste* unlocks plastic waste's economic value while aligning profit with environmental benefit. *Pilot Region:* The project will pilot in Bogotá, Colombia, a city of 10 million with no deposit-return system and a recycling rate under 10%(United Nations Environment Programme 2023). Bogotá generates 280,000 tonnes of plastic waste annually, most of which is landfilled(United Nations Environment Programme 2023). This pilot leverages high waste generation to demonstrate the impact of incentives, with potential for scaling across Colombia and LAC markets.

Market Definition and Size: The target market for *Money for Waste* can be defined on three levels (see Table 0.1). **Total Addressable Market (TAM):** This represents the global potential of monetizing plastic waste. The worldwide plastic packaging waste stream (over 120 million tonnes annually) has a lost material value of $80–120 billion(Ezeudu, Tenebe, and Ujah 2024), making the TAM €90 billion per year (assuming 1 USD ≈ 0.9 EUR). This reflects the untapped economic value of recyclable plastics in a linear economy(Ezeudu, Tenebe, and Ujah 2024). **Serviceable Available Market (SAM):** This is the portion of the TAM that *Money for Waste* could realistically target in its initial expansion. In LAC emerging markets, where recycling infrastructure is underdeveloped, the SAM is €8 billion annually. LAC generates 18 million tonnes of plastic waste per year (with <10% recycled)(United Nations Environment Programme 2023); monetized at €500 per tonne, this yields €9 billion. **Serviceable Obtainable Market (SOM):** This is the share of the SAM the startup can capture in the near term. Initially, *Money for Waste* will target Bogotá. Colombia's plastic waste (1.3 million tonnes/year) corresponds to €0.5 billion in value. Bogotá, with 0.28 million tonnes/year, offers an SOM of €120 million per year (at €430/tonne). Realistically, not all of this can be captured.

Year 1 Sales Projection: The pilot is projected to achieve significant results in its first year, capturing 5% of the serviceable market in Bogotá through early adopters and partnerships. Key figures are summarized in Table 0.3.

Statistic	**Value**
Global plastic production (1950–2015)	8.3 billion tonnes
Global plastic waste generated	6.3 billion tonnes
Global recycling rate	9%
Plastic packaging recycling rate	14%
Annual economic loss	\$80–120 billion
LAC recycling rate	Less than 10%
LAC daily plastic waste in open dumps	17,000 tonnes

Figure 0.1: Key statistics on global and regional plastic waste challenges.

Table 0.1: Market size estimates for *Money for Waste* (in euros). TAM is global; SAM is the Latin American market; SOM is initially focused on Bogotá, Colombia.

Market Segment	**Scope and Assumptions**	**Annual Size**
TAM (Global)	All plastic waste worldwide (esp. packaging). ~300 Mt/year generated; lost material value ~ \$100 B.	€90 B
SAM (LAC Region)	Plastic waste in Latin America (18 Mt/year not recycled). Assumes average €500/tonne value if recycled.	€8 B
SOM (Bogotá pilot)	Plastic waste in Bogotá (0.28 Mt/year). Assumes €430/tonne average value; initial obtainable share ~28,000 t (10%).	€12 M

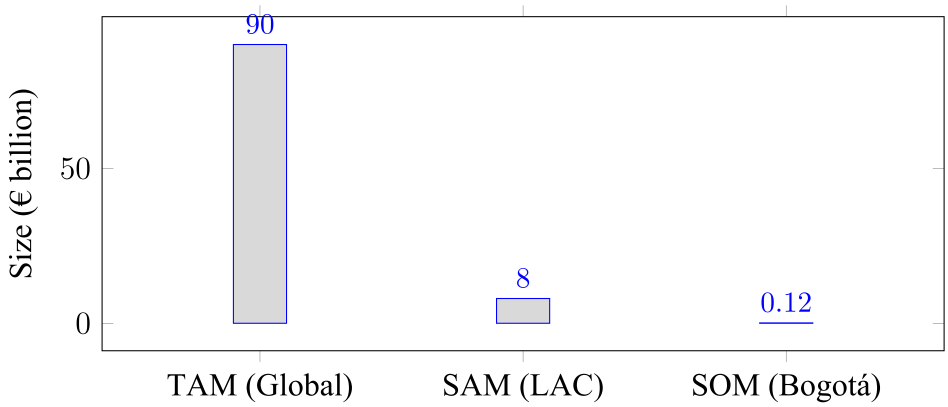

Figure 0.2: Market Size by Segment (in € billions).

Metric	**Value**
Market capture rate	5%
Plastics collected	14,000 tonnes
Average revenue per tonne	€500
Year 1 sales projection	€7 million

Figure 0.3: Year 1 sales projection for the Bogotá pilot.

Bibliography

Ezeudu, Obiora B., Imokhai T. Tenebe, and Chika O. Ujah (2024). “Status of Production, Consumption, and End-of-Life Waste Management of Plastic and Plastic Products in Nigeria: Prospects for Circular Plastics Economy”. In: *Sustainability* 16.18, p. 7900. DOI: 10.3390/su16187900.

Geyer, Roland, Jenna R. Jambeck, and Kara L. Law (2017). “Production, use, and fate of all plastics ever made”. In: *Science Advances* 3.7, e1700782. DOI: 10.1126/sciadv.1700782.

Husaini, Danladi C. et al. (2024). “Plastic Waste in Latin America and the Caribbean (LAC): Impact on the Environment and Public Health—A Systematic Review”. In: *Journal of Toxicology* 2024, p. 5698516. DOI: 10.1155/2024/5698516.

United Nations Environment Programme (2023). *Blue Awakening as Latin American and Caribbean states say no to plastic*. https://www.unep.org/news-and-stories/story/blue-awakening-latin-american-and-caribbean-states-say-no-plastic. Accessed: May 1, 2025.